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DP-CopulaGAN: A Differentially Private Synthetic Data Generator Using Copulas and Generative Adversarial Networks

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Software

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Summary

The increasing demand for data sharing in research, government, and industry has heightened the need for methods that protect individual privacy while retaining analytical usefulness. Synthetic data generation—creating artificial datasets statistically similar to real data—offers a promising solution but often encounters two major challenges: (1) preserving complex dependency structures in tabular data, and (2) providing formal, mathematically guaranteed privacy protection.

`DP-CopulaGAN` is a Python package designed to meet these challenges. It integrates **Gaussian copula transformations** with a **Wasserstein Generative Adversarial Network (WGAN-GP)** trained using **Differentially Private Stochastic Gradient Descent (DP-SGD)**. This combination enables the generation of high-utility synthetic tabular data under formal (ε, δ)-differential privacy guarantees.

The software is intended for users in **official statistics**, **healthcare**, **machine learning research**, and **privacy engineering**, especially where sensitive microdata cannot be released directly. `DP-CopulaGAN` supports both **conditional** (with labels) and **unconditional** (unlabeled) generation and includes built-in evaluation metrics for statistical similarity and machine-learning utility.

Statement of Need

Organizations that collect sensitive microdata—national statistical offices, hospitals, financial institutions, and research agencies—must balance public data access with privacy protection. Traditional disclosure control methods such as suppression, top-coding, noise injection, or aggregation often degrade data utility, limiting their value for scientific research and policy analysis (Hundepool et al., 2012).

Synthetic data provides an alternative, but major gaps remain:

- Many synthetic data tools **do not provide formal privacy guarantees**. This exposes them to attacks such as membership inference (Shokri et al., 2017).
- Differentially private generative models exist, but they typically **struggle to preserve correlations and non-Gaussian distributions**, especially in mixed-type tabular data.
- Copula-based models preserve dependencies well, but most are **not private**.

`DP-CopulaGAN` addresses these limitations by combining **dependency-preserving copulas** with **differentially private GAN training**, enabling the release of useful synthetic data with mathematically guaranteed privacy.

The package is designed for:

1. **Official statistics agencies** generating synthetic public-use microdata (Drechsler & Reiter, 2011; Reiter, 2005).
2. **Healthcare and biomedical researchers** needing privacy-preserving patient-level datasets (R. J. Chen et al., 2021).
3. **Machine learning practitioners** training models on private data.
4. **Researchers and students** studying differential privacy, synthetic data, or generative modeling.

Software Description

`DP-CopulaGAN` provides a complete, production-ready framework for private synthetic data generation.

Key Features

- **Gaussian Copula Transformation:** Models complex dependencies and transforms arbitrary marginals into a Gaussian latent space (Nelsen, 2006).
- **WGAN-GP Generator and Critic:** Ensures stable GAN training using gradient penalty (Gulrajani et al., 2017).
- **Differential Privacy via DP-SGD:** Applies gradient clipping and Gaussian noise during critic updates, yielding formal privacy guarantees (Abadi et al., 2016).
- **Conditional and Unconditional Modes:** Supports labeled datasets for supervised tasks and unlabeled datasets for general tabular synthesis.
- **Comprehensive Evaluation Tools:** Statistical metrics (JSD, Wasserstein, correlation RMSE) and ML utility metrics (TSTR with 12 classifiers, multiclass F1, and unsupervised utility scores).
- **Automatic Preprocessing:** Type detection, categorical encoding/decoding, NaN handling, and data validation.

Architecture Overview

1. **Preprocessing:** Detects variable types and encodes categorical labels if present.
2. **Copula Modeling:** Learns a Gaussian copula representation either globally (unconditional) or per class (conditional).
3. **GAN Training:** Uses WGAN-GP with spectral normalization (Miyato et al., 2018) and DP-SGD to ensure privacy.
4. **Sampling:** Generates synthetic latent samples, transforms them back using the inverse copula, and decodes categorical variables.
5. **Evaluation:** Computes statistical similarity and supervised/unsupervised ML utility.

Example Usage

```python
from dp_copulagan import DPCopulaGAN
import pandas as pd

data = pd.read_csv("sensitive.csv")

model = DPCopulaGAN(
    epsilon=1.0,
    delta=1e-5,
    label_col="income"
)

model.fit(data)
synthetic = model.sample(10000)
```

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State of the Field

Synthetic data generation spans both **classical statistical approaches** and **modern deep generative models**. Early approaches include CART-based synthesis and multiple imputation (Drechsler & Reiter, 2011; Reiter, 2005), which perform well for simple structures but struggle with high-dimensional or non-Gaussian tabular data.

The **Synthetic Data Vault (SDV)** ecosystem (Patki et al., 2016) introduced widely used neural models such as **CTGAN** (Xu et al., 2019), **TVAE**, and **CopulaGAN**. These models generate high-utility synthetic data but lack **formal differential privacy**, making them unsuitable for sensitive datasets.

Several **differentially private generative models** have been proposed:

- **DP-WGAN** (Xie et al., 2018)
- **PATE-GAN** (Jordon et al., 2019)
- **DP-VAE** (T. Chen et al., 2018)

While these models offer privacy guarantees, they typically struggle with tabular data because of **non-Gaussian marginals**, **mixed feature types**, and **complex dependency structures**. As a result, they often fail to preserve realistic multivariate relationships.

Other tabular GAN approaches—such as **TableGAN** (Park et al., 2018) and **TGAN** (Xu & Veeramachaneni, 2018)—produce high-utility synthetic data but do not include differential privacy. **DP-CTGAN** (Torfi et al., 2022) incorporates privacy but does not use copula transformations, which limits its capability to preserve correlation structures.

DP-CopulaGAN advances the field by integrating:

1. **Gaussian copula modeling** for preserving multivariate dependencies
2. **Differentially private GAN training** for formal privacy guarantees
3. **Tabular-data-optimized design**, including conditional and unconditional pipelines and mixed-type variable handling

This combination significantly improves both **privacy protection** and **statistical fidelity**, bridging the gap between high-utility synthetic data generation and strong differential privacy guarantees.

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